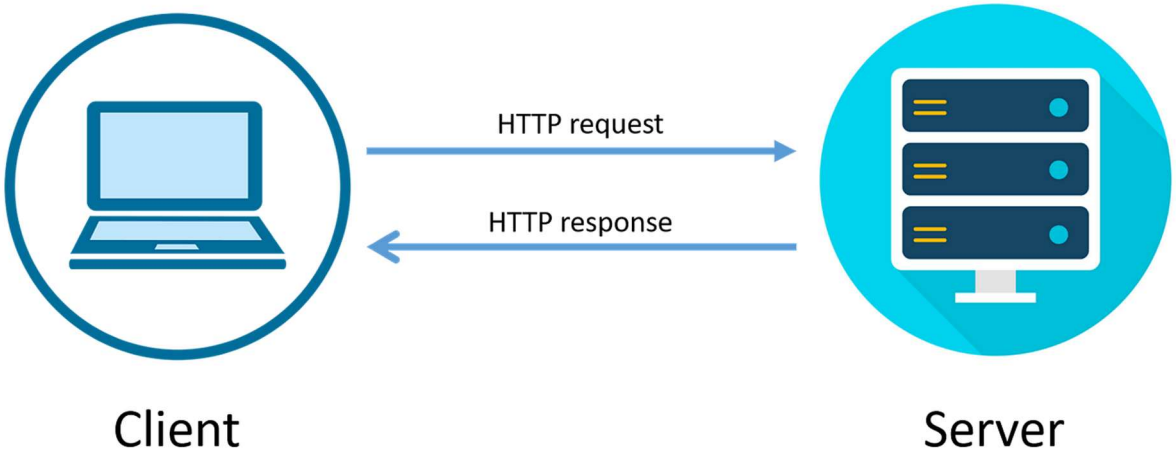
Assignment Name

Internet Explorer

Description

Draw and explain the client server architecture and trace what happens when you open a website



Introduction

The internet functions through continuous communication between user devices and remote systems. A website does not exist inside your computer. A website is stored on another machine connected through the internet. When you open a website, your device sends a request and receives data in return. This interaction follows a structured design known as client server architecture.

Client server architecture forms the foundation of web browsing, online services, and cloud platforms. Every action performed on the web follows this model. Understanding this concept helps you learn how browsers, servers, and networks work together to deliver information.

Client Server Architecture

Client server architecture is a distributed computing model. One system requests a service. Another system provides the service. Each system performs a defined role.

Client

A client is the end user device. Examples include desktop computers, laptops, smartphones, and tablets. The web browser installed on the device acts as the client. The client initiates communication. The client sends requests and receives responses.

Server

A server is a high performance computer. It stores websites, applications, and data. The server listens for incoming requests from multiple clients. It processes each request and sends appropriate responses.

Network

The network connects the client and the server. The internet consists of routers, cables, switches, and transmission protocols. This network ensures data travels between devices accurately and securely.

Working principle

The client sends a request
The server receives the request
The server processes the request
The server sends a response
The client displays the response

This separation improves efficiency, security, and scalability. Many users access the same server at the same time without interference.

Types of Servers Involved

Web Server

A web server handles browser requests. It delivers static content such as HTML pages, images, and stylesheets.

Application Server

An application server runs backend logic. It handles user input, authentication, and business rules. It generates dynamic content.

Database Server

A database server stores structured data. Examples include user records, login details, and transaction history.

DNS Server

A DNS server converts website names into numerical IP addresses. This step allows devices to locate servers on the internet.

Each server performs a specific role to ensure smooth website operation.

Client Server Architecture Diagram Explanation

In a typical diagram, the client appears on one side and the server appears on the other side. The client connects through the internet. Arrows indicate request and response flow.

The request arrow shows data moving from the client to the server.
The response arrow shows data moving from the server to the client.

This visual representation helps understand how communication flows between systems.

What Happens When You Open a Website

Step 1. Entering the website address

You type a website address into the browser address bar. This address is called a URL. The browser begins the process of locating the website server.

Step 2. Browser cache check

The browser checks stored information. If the website was visited earlier, saved records may exist. This reduces loading time.

Step 3. DNS resolution

If no record exists, the browser contacts a DNS server. The DNS server translates the website name into an IP address. This IP address identifies the server location on the internet.

Step 4. Establishing a connection

The browser creates a connection with the server using the IP address. This step ensures reliable communication between both systems.

Step 5. Secure communication setup

If the website uses secure communication, encryption is established. This protects data exchanged between the browser and the server.

Step 6. Sending the request

The browser sends an HTTP request to the server. The request contains the page name and required information.

Step 7. Server processing

The server receives the request.
The server checks permissions.
Backend logic executes if required.
The server retrieves data from databases if needed.

Step 8. Server response

The server sends an HTTP response back to the browser. The response includes the requested webpage content and a status message.

Step 9. Page rendering

The browser reads the response.
HTML builds the structure.
CSS applies design.
JavaScript enables interaction.

Step 10. Loading additional resources

Images, fonts, and scripts load through separate requests. The page becomes fully visible after all resources load.

Advantages of Client Server Architecture

Centralized data storage improves consistency
Security rules apply at the server level
Maintenance becomes easier
Performance improves with caching
Multiple users access services simultaneously

This architecture supports modern web applications and cloud systems.

Real Life Example

You open an online banking website.
Your browser requests the login page.
The server sends the page layout.
When you enter credentials, the application server validates them.
The database server retrieves account data.
The server sends updated information to your screen.

Every step follows the client server model.

Conclusion

Client server architecture supports all internet activities. Every website request follows a structured communication process. Understanding this model builds strong fundamentals in networking and web technologies. This knowledge supports learning in web development, cloud computing, and cybersecurity.